FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

Humayun Arshad

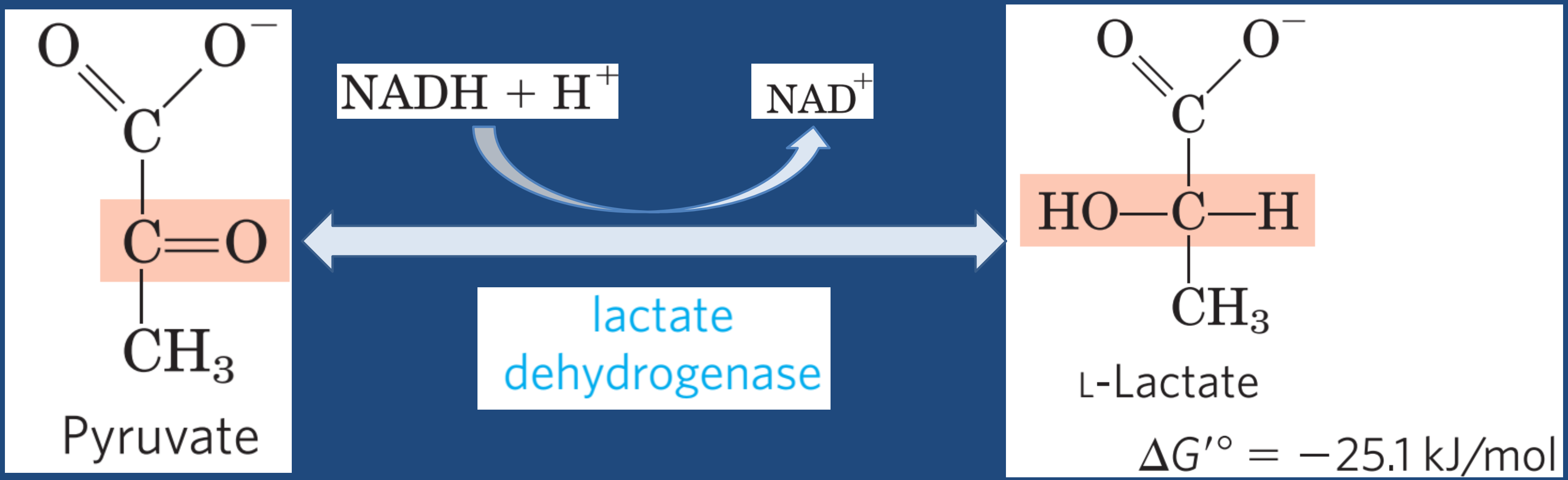
FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

- Under aerobic conditions, the pyruvate formed in the final step of glycolysis is oxidized to acetate (acetyl-CoA), which enters the citric acid cycle and is oxidized to CO2 and H2O. The NADH formed by dehydrogenation of glyceraldehyde 3-phosphate is ultimately reoxidized to NAD^+ by passage of its electrons to O_2 in mitochondrial respiration.
- Under hypoxic (low-oxygen) conditions, however—as in very active skeletal muscle, in submerged plant tissues, in solid tumors, or in lactic acid bacteria—NADH generated by glycolysis cannot be reoxidized by O_2.
- Failure to regenerate NAD^+ would leave the cell with no electron acceptor for the oxidation of glyceraldehyde 3-phosphate, and the energy-yielding reactions of glycolysis would stop.
- NAD^+ must therefore be regenerated in some other way.

FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

1. Lactic Acid Fermentation

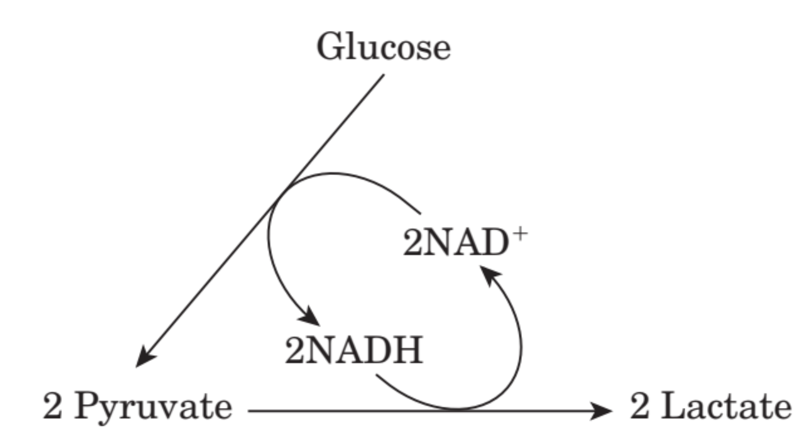
- When animal tissues cannot be supplied with sufficient oxygen to support aerobic oxidation of the pyruvate and NADH produced in glycolysis, NAD^+ is regenerated from NADH by the reduction of pyruvate to **lactate**.



FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

1. Lactic Acid Fermentation

- Because the reduction of two molecules of pyruvate to two of lactate regenerates two molecules of NAD^+, there is no net change in NAD^+ or NADH.



FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

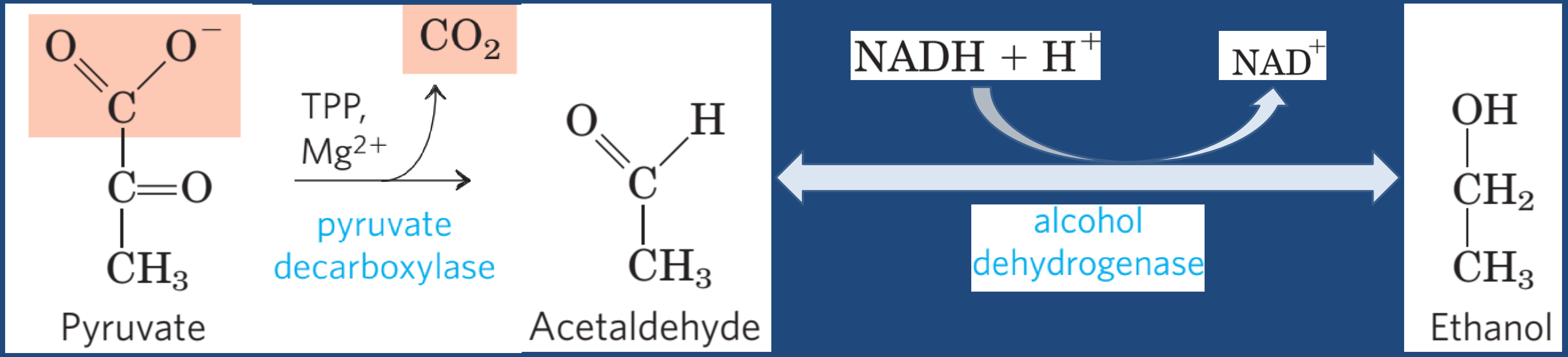
1. Lactic Acid Fermentation

- The lactate formed by active skeletal muscles (or by erythrocytes) can be recycled; it is carried in the blood to the liver, where it is converted to glucose during the recovery from strenuous muscular activity.
- When lactate is produced in large quantities during vigorous muscle contraction (during a sprint, for example), the acidification that results from ionization of lactic acid in muscle and blood limits the period of vigorous activity.
- **Fermentation** is the general term for such processes, which extract energy (as ATP) but do not consume oxygen or change the concentrations of NAD^+ or NADH.

FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

2. Alcoholic Fermentation

- Yeast and other microorganisms ferment glucose to ethanol and CO_2, rather than to lactate.



FERMENTATION: THE ANAEROBIC FATE OF PYRUVATE

2. Alcoholic Fermentation

- Pyruvate decarboxylase requires Mg^{2+} and has a tightly bound coenzyme, thiamine pyrophosphate.
- Ethanol and CO_2 are thus the end products of ethanol fermentation, and the overall equation is:

$$\text{Glucose} + 2\text{ADP} + 2\text{P}_i \rightarrow 2\ \text{ethanol} + 2\text{CO}_2 + 2\text{ATP} + 2\text{H}_2\text{O}$$

- The CO_2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characteristic carbonation of champagne.
- In baking, CO_2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sugar causes dough to rise.